

RAJ KUMAR TIWARI

Software Developer

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SUMMARY

Software Engineer specializing in **Frontend** development (**2+ years**) using **React.js, Redux, Next.js, JavaScript, TypeScript**, and **Material UI**. Possessing intermediate expertise in **Data Structures & Algorithms** and **CPP**, and strong proficiency in data persistence with **MongoDB**. Highly motivated to leverage foundational knowledge in **Backend** technologies (**Python, FastApi, Node.js, Express.js, WebSockets**) and advanced data solutions, including **Retrieval-Augmented Generation (RAG)**, **Langchain** and **OpenAI** integration.

TECHNICAL SKILLS

- **Backend:** Python, FastApi, Node.js, Express.js, REST APIs, WebSockets
- **Frontend:** React.js, Redux, Next.js, JavaScript, TypeScript, Material UI
- **Databases:** MongoDB
- **Cloud & DevOps:** Git
- **Others:** Data Structures & Algorithms, Retrieval-Augmented Generation (RAG), Langchain, CPP, OpenAI

WORK EXPERIENCE

Software Engineer

Jan. 2024 - Apr. 2026

- **Frontend Expertise (2+ Years):** Built responsive and high-performance user interfaces using **React.js, Redux, TypeScript, JavaScript**, and **Material-UI**, with a strong focus on scalability and user experience. Implemented API integration using **Redux** for centralized state management, ensuring efficient data handling and optimized application performance.
- **Backend Fundamentals:** Developed foundational backend skills by building **APIs** and **server-side logic** using **Node.js** and **Express.js**, along with database integration using **MongoDB**
- **AI Integration Experience:** Developed and integrated **OpenAI-powered LLM APIs** into full-stack applications, enabling intelligent features such as real-time responses and enhanced user interaction.
- Implemented **Web Workers** to handle computationally intensive tasks, improving UI responsiveness and preventing main thread blocking.

Software Engineer Intern

Sep. 2023 - Dec. 2023

- Gained hands-on experience in frontend development using **React.js, TypeScript, JavaScript, Redux**, and **Material-UI**, building reusable UI components and understanding scalable architecture
- Explored backend development fundamentals, including **API handling, data flow**, and **frontend-backend integration**, strengthening full-stack development concepts

PROJECTS

Review Portal Project

Developed and deployed a full-stack Review Portal application by building both frontend and backend, along with integrating OpenAI APIs for AI-powered functionality. There are 2 portals.

- **User Portal :** Designed and implemented a user-facing portal using **React.js, TypeScript, Material-UI, Redux**, and **Local Storage**, where users can generate reviews using **OpenAI**, submit those reviews, and upload screenshots as proof of submission.
- **Admin Portal :** Built an admin dashboard using **React.js, TypeScript, Material-UI**, and **Redux** to manage and moderate user submissions by reviewing uploaded screenshots and approving or rejecting the reviews.

Trading View Project

Developing a modern trading platform using **Next.js**, focusing on building a high-performance and scalable frontend with a seamless user experience. Currently implementing dynamic UI components and real-time data visualization features, with plans to extend the project by developing robust backend services and APIs.

CheckIn Project

Developed a cross-platform mobile application using **Flutter** and **Dart** for efficient check-in and check-out management . Implemented a lightweight local JSON-based database for data handling and offline support

AI Agent with RAG (Retrieval-Augmented Generation)

Developed an intelligent AI agent using **LangChain** that leverages multiple data sources and tools to provide context-aware and accurate responses.

- Built a **RAG pipeline** by loading web-based documents, splitting text, generating embeddings, and storing them in a **FAISS vector database** for efficient retrieval.
- Integrated external data sources like **Wikipedia API** and **Arxiv API** to enhance knowledge coverage.
- Created a **custom retriever tool** and enabled dynamic tool selection using a **tool-calling agent architecture**.
- Utilized **ChatOllama (Llama 3.1)** as the core LLM for generating responses.
- Implemented an **Agent Executor** to process user queries and fetch relevant information from multiple sources in real time.

EDUCATION

B.tech in Computer Science and Engineering (AI,ML)	2020 - 2024
Galgotias University Greater Noida Uttar Pradesh (GU)	